

Sequences — 5.1

Definition: A sequence, denoted (a_n) or $\{a_n\}$ or $\{a_n\}_{n=1}^{\infty}$ is an infinite ordered list of real numbers. We call n the index variable. Alternatively, a sequence is a function with domain the natural numbers.

We may also write sequences as lists $a_1, a_2, a_3, \dots$ or $(a_1, a_2, a_3, \dots)$

Examples of sequences:

(a) $(1, 1, 1, 1, 1, \dots)$

$$a_n = 1$$

(b) $(1, 2, 3, 4, 5, \dots)$

$$a_n = n$$

(c) $(\pi/2, \sin(1), 1, e^{2001}, \dots)$

$$a_n = ?$$

(d) $(1, 1, 2, 3, 5, 8, 13, \dots)$

$$a_n = a_{n-1} + a_{n-2}$$

(e) $(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots)$

$$a_n = \frac{1}{n}$$

(f) $(\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots)$

$$a_n = \frac{n}{n+1}$$

(g) $(1, -1, 1, -1, 1, \dots)$

$$a_n = (-1)^{n-1}$$

(h) $(3, 3.1, 3.14, 3.141, \dots)$

$$a_n = \frac{\lfloor 10^{n-1} \pi \rfloor}{10^{n-1}}$$

$$\frac{\lfloor 10^{3-1} \pi \rfloor}{10^{3-1}} = \frac{\lfloor 314.1592 \rfloor}{10^2} = \frac{314}{100} = 3.14$$

We can define a sequence recursively by specifying an initial value/values and writing a rule to determine the next value in terms of the previous values.

$$a_1 = 1$$

$$a_2 = 1$$

$$a_3 = 2$$

$$a_4 = 3$$

$$a_5 = 5$$

$$a_6 = 8$$

$$a_7 = 13$$

⋮

$$a_1 = 1$$

$$a_2 = 1$$

$$a_n = a_{n-1} + a_{n-2}$$

Fibonacci sequence

Example : $a_n = n! = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1$

$$(1, 2, 6, 24, 120, 720, \dots)$$

Example : $a_n = \frac{e^n}{n!}$